

Identifying Data Requirements for an Online Shopping Platform

Primary Key

A Primary Key (PK) is an attribute that uniquely identifies every record in a database table. It helps prevent duplicate records and ensures that every piece of information can be retrieved accurately. A primary key cannot contain duplicate or NULL values and is essential for maintaining relationships between different entities in a database.

Example:

Every order placed by a customer is assigned a unique Order_ID such as ORD1001. Even if two customers place orders at the same time, each order will have a different Order_ID.

Core Entities

1. Customer

Stores customer registration and contact details.

Attributes

Customer_ID (Primary Key)

Customer_Name

Email

Mobile_Number

Password

Address

Date_of_Registration

2. Product

Stores details of products available on the shopping website.

Attributes

Product_ID (Primary Key)

Product_Name

Brand

Price

Stock_Quantity

Product_Image

Description

3. Category

Organizes products into different sections.

Attributes

Category_ID (Primary Key)

Category_Name

Category_Type

4. Supplier

Stores information about suppliers providing products.

Attributes

Supplier_ID (Primary Key)

Supplier_Name

Contact_Number

Email

Supplier_Address

5. Delivery Address

Stores multiple addresses of customers.

Attributes

Address_ID (Primary Key)

Customer_ID

House_No

Street

City

State

Pincode

6. Order

Stores order information.

Attributes

Order_ID (Primary Key)

Customer_ID

Order_Date

Order_Status

Total_Amount

7. Payment

Stores payment details.

Attributes

Payment_ID (Primary Key)

Order_ID

Payment_Mode

Payment_Status

Payment_Date

8. Product Review

Stores customer opinions about purchased products.

Attributes

Review_ID (Primary Key)

Customer_ID

Product_ID

Rating

Review_Message

Additional Features

Search

Used to search products using keywords.

Attributes

Search_ID

Search_Keyword

Search_Date

Filter

Allows customers to narrow search results.

Filter Options

Brand

Category

Price Range

Customer Rating

Discount

Availability

Size

Sort

Displays products in different orders.

Sort Types

Price (Low to High)

Price (High to Low)

Latest Arrivals

Best Rated

Most Popular

Gender

Used mainly for fashion and lifestyle products.

Options

Men

Women

Kids

Unisex

Notifications

Stores alerts sent to customers.

Attributes

Notification_ID

Customer_ID

Message

Date

Status (Read/Unread)

Order Tracking

Stores delivery progress.

Attributes

Tracking_ID

Order_ID

Courier_Name

Current_Location

Expected_Delivery_Date

Delivery_Status

Conclusion

An Online Shopping Platform requires multiple entities to store customer, product, supplier, category, payment, and order information efficiently. Additional features such as search, sorting, filtering, notifications, gender classification, delivery addresses, and order tracking enhance the shopping experience. The use of Primary Keys ensures that every record is unique and supports accurate relationships among all database tables.